Virtualization

Objectives:

At the end of this episode, I will be able to:

1. Identify components of virtualization and determine each components purpose.

Exam Objective: *2.5 - Summarize the purpose and operation of virtualization.*

Description: In this episode, the viewer will identify the purpose of various virtualization technologies such as virtual networking components, resource allocation, provisioning and management.

- What is virutalization?
- What is a hypervisor? (Type 1 vs. Type 2, scalability)
- Management interfaces for virtual machines
    - GUI-based application
    - Browser-based interface
    - Commandline/terminal/shell
- Provision a Guest
- Host vs. guest
    - Guest vs. Virtual Machine, which is it?
- Identify host resources
    - MSInfo/System Information (Total)
    - Task Manager and Resource Manager
- Provision a guest machine and allocate resources
    - Use Hyper-V to create a VM called websrv01
        - Name: websrv01
        - Generation 2
        - Memory allocation: 4096
        - Virtual Network: Default (performing NAT)
        - ``` Three vSwitch types : Bridged (Direct Access)or through NAT, host-to-guest, guest-to-guest

      ```
        - Storage Drive: 60 GB
        - Installation Options: Install OS from a bootable image
- Verify successful guest machine
    - Review settings
    - Connect to VM and start

---

### External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Not applicable if blank
      ```